Adding, Subtracting, and Multiplying Polynomials

Adding and subtracting by like terms, multiplying with an area box, and checking a product by substitution or by factoring back

Directions:

- Sums and differences come with a **like-terms table**: one column per power. Fill the rows, then combine down each column — only terms in the same column may be combined.
- Products come with an **area box**: one row per term of the first factor, one column per term of the second. Every cell is one multiplication, and the answer is the sum of the cells.
- Write the final polynomial in descending order of degree on the answer line.

1. Add: $(3x^3 + 2x^2 - 5) + (x^3 - 4x^2 + 7)$

	x^3	x^2	x	**constant**
First	$3x^3$	$2x^2$		-5
Second	x^3	$-4x^2$		7
Sum				

Answer: ________________

2. Add: $(2x^4 - x^2 + 6x) + (5x^4 + 3x^2 - 6x)$. Notice what happens in one of the columns and say what that tells you about the degree of the answer.

	x^4	x^2	x	**constant**
First	$2x^4$	$-x^2$	$6x$	
Second	$5x^4$	$3x^2$	$-6x$	
Sum				

Answer: ________________

3. Subtract: $(4x^3 - 2x + 9) - (x^3 + 5x - 3)$. Write the second row as it looks *after* the minus sign has been distributed to every term.

	x^3	x	**constant**
First	$4x^3$	$-2x$	9
Second, negated			
Difference			

Answer: ________________

4. Multiply: $(x+3)(x^2-2x+5)$. Fill every cell of the area box, then add the cells.

$\times$	x^2	$-2x$	5
x			
3			

Answer: ____________________

5. Subtract: $(5x^4+3x^3-x)-(2x^4-x^3+4x-8)$. Two of the signs inside the second polynomial are already negative — be careful what the distributed minus does to them.

	x^4	x^3	x	**constant**
First	$5x^4$	$3x^3$	$-x$	
Second, negated				
Difference				

Answer: ____________________

6. Checking a product by substitution. Problem 4 claimed $(x+3)(x^2-2x+5) = x^3 + x^2 - x + 15$. Substitute $x = 2$ into the *expanded* form and record the value; then substitute $x = 2$ into the factored form and confirm the two agree.

Form	**Value at** $x = 2$
$(x+3)(x^2-2x+5)$	
$x^3 + x^2 - x + 15$	

Answer: ________________

7. Multiply: $(2x^2 - 3)(x^2 + 4x - 1)$. Two of the six cells give like terms — circle them before you add.

$\times$	x^2	$4x$	-1
$2x^2$			
-3			

Answer: ________________

8. Checking a product by factoring back. Expanding $(x-2)(x+2)(x+1)$ is claimed to give x^3+x^2-4x-4. Factor x^3+x^2-4x-4 by grouping and state whether the claim holds.

Grouping step	**Result**
Group the first two and last two terms	
Factor each group, then the common factor	

Answer: ________________

9. Multiply: $(x^2+x-1)^2$. Write the square as a product of two identical trinomials first, then use a three-by-three area box.

$\times$	x^2	x	-1
x^2			
x			
-1			

Answer: ________________

10. A student writes $(x^3-2)^2 = x^6 - 4$. Fill in the area box to show every cell the student left out, explain in one sentence which rule was misapplied, and write the correct expansion.

$\times$	x^3	-2
x^3		
-2		

Answer: __________________